

Universal Report Analyzer

Installation & Complete Workflow Guide

1. Project Setup

This section explains how a new developer can clone, configure and run the application locally.

Step 1: Clone the Repository

```
git clone https://github.com/abdulsanad27/Universal-report-generator.git
cd Universal-report-generator
```

Downloads the latest source code from GitHub and moves into the project directory.

Step 2: Create Virtual Environment

```
python3 -m venv venv
```

Creates an isolated Python environment so project dependencies do not affect the system Python.

Step 3: Activate Virtual Environment

```
source venv/bin/activate
```

Activates the virtual environment. All Python packages will now be installed only inside this project.

Step 4: Install Dependencies

```
pip install -r requirements.txt
```

Installs every library required by the application such as Flask, Jinja2, BeautifulSoup and other dependencies.

Step 5: Run the Application

```
python3 universal_report_analyzer.py
```

Starts the Flask web server locally.

Step 6: Verify Server

```
Running on http://127.0.0.1:5000
```

This confirms that the application has started successfully.

Step 7: Open Browser

`http://127.0.0.1:5000`

Open the application in your browser.

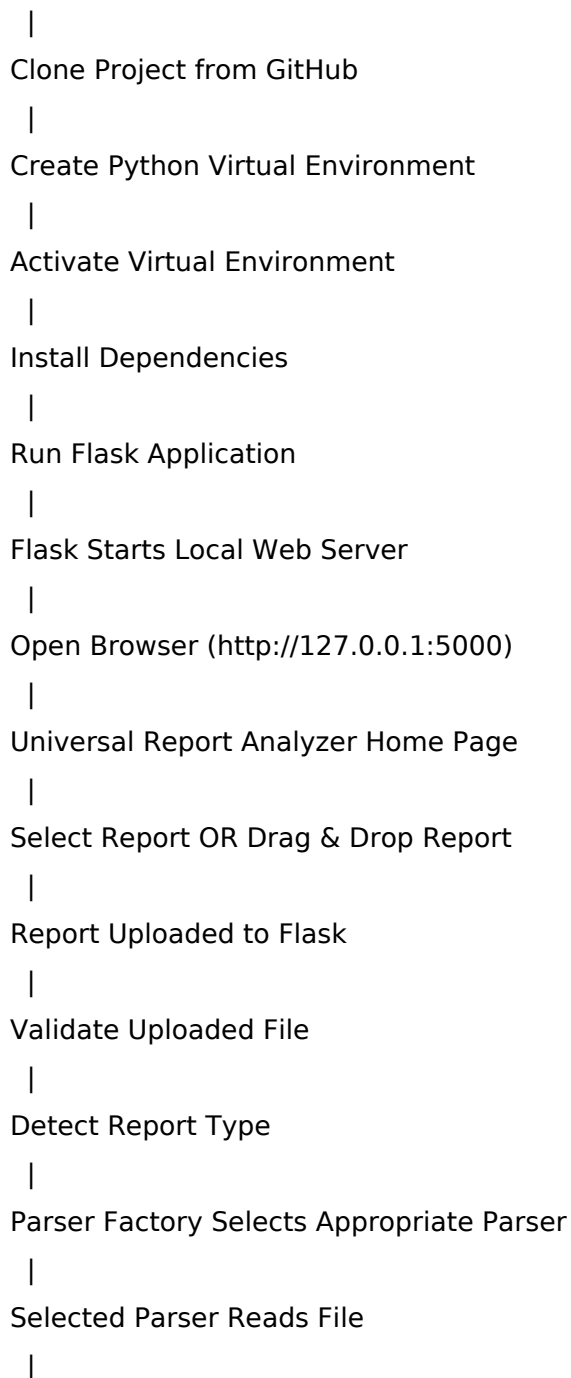
Step 8: Generate Report

Upload `sample_test.xml` and click Generate Report.

The application processes the report and displays a readable HTML dashboard.

2. End-to-End Application Workflow

START



Extract:

- Class Name
- Test Method
- Status
- Execution Time
- Failure Message

|

Convert to Universal Report Model

|

Calculate Statistics

- Total Tests
- Passed
- Failed
- Skipped
- Execution Time

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Generate HTML Report

(Jinja2 + HTML + CSS + JavaScript)

|

Display Interactive Dashboard

|

Review Report

|

Save Page As...

|

Share BOTH:

- report.html
- report_files folder (contains CSS & JS)

|

END

3. Important Notes

- Always activate the virtual environment before running the application.
- If a new library is added, update requirements.txt using:
pip freeze > requirements.txt
- When sharing the generated report, send BOTH the HTML file and the report_files folder.
Without the CSS and JavaScript files, the report will lose its styling and interactive features.
- The application automatically detects the uploaded report type and selects the correct parser.

4. Troubleshooting

Application does not start:

- Verify the virtual environment is activated.
- Reinstall dependencies using `pip install -r requirements.txt`.

ModuleNotFoundError:

- Activate the venv.
- Install missing packages.

Blank UI after sharing HTML:

- Ensure both `report.html` and `report_files` folder are shared together.